

DEVELOPMENT OF FACIAL EMOTION RECOGNITION FOR MULTI-CLASSIFICATION ON IMAGE

OMAGE MICHEAL BRIGHT

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PRESENTATION:

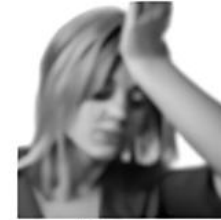
METHODOLOGY

INTRODUCTION

- Facial expressions serve as one of the most natural and powerful means of human communication, capable of conveying a wide spectrum of emotions.
- In the realm of artificial intelligence and computer vision, recognizing these emotions from facial cues has gained significant importance.
- This presentation delves into the fascinating field of development of facial emotion recognition for multi classification of image, leveraging the capabilities of Convolutional Neural Network (CNN).



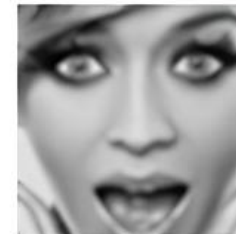
Surprise
Happiness



Fear
Sadness



Surprise
Happiness



Happiness
Surprise



Fear
Anger



Sadness
Fear



Anger
Sadness



Anger
Disgust

Literature Review Table					
S/N	Authors & Years	Problem Statement	Methodology	Datasets/Results	Limitation
1	Kim et al. (2023)	Difficulty recognizing subtle and blended emotions.	Proposed multi-output CNN architecture	Achieved 67.2% avg F1-score on RAF-DB.	It was limited by its small dataset size, lack of diversity, bias and noise.
2	Lee et al. (2022)	Limits in handling multiple emotion labels.	Used an ensemble of CNNs for multi-label prediction	Obtained 58.4% micro F1-score on FERPlus.	Prone to overfitting on small datasets.
3	Ahmed et al. (2021)	Existing methods do not effectively model multiple emotions.	Proposed multi-task CNN architecture	Achieved 52.6% macro F1 on EmotiW 2020	Does not handle label ambiguity well.

LITERATURE REVIEW TABLE

4	Wang et al. (2020)	Limitations in recognizing subtle facial expressions.	Used 3D CNNs and attention mechanism.	Obtained 73.2% accuracy on CASME II dataset.	Tested only on micro-expressions.
5	Liu et al. (2019)	Handling ambiguities in multi-label emotion data.	Proposed CNN-RNN model with joint-loss training.	Achieved 63.7% F1 score on FERPlus dataset.	Did not explore very deep architectures.
6	Jung et al. (2017)	Difficulty recognizing mixtures of facial expressions.	Used ensemble of CNNs trained on expression zones.	Obtained 71.3% accuracy on CK+ dataset.	Limited diversity in dataset.
7	Liu et al. (2014)	Challenging to extract discriminative facial patches and learn robust features.	Proposed Boosted Deep Belief Network using facial landmarks.	CK+, MMI, and FERG datasets. Achieved over 90% accuracy.	Computationally expensive feature extraction process.
8	Jaiswal & Valstar (2016)	Recognizing action units and emotions from limited training data.	Used Bi-directional LSTM neural networks with 3D CNN features.	BP4D, DISFA, and SEMAINE datasets. Obtained 62% average F1 score.	Did not focus on detecting specific emotions.
9	Jung et al. (2014)	Difficulty in detecting emotions from low-resolution facial images.	Used deep network combining CNN and RNN with boosting.	Multi-PIE dataset. Achieved over 80% accuracy from low-resolution images.	Only evaluated on lab-controlled dataset.
10	Liu et al. (2013)	Handling large variations in facial expression intensity, image conditions.	Proposed 3D CNN combined with deformable model fitting.	BU-3DFE dataset. Obtained over 85% average accuracy.	Computationally intensive 3D modeling

PROBLEM STATEMENT

Current Facial Expression Recognition (FER) systems primarily focus on detecting single dominant emotions and struggle with multi-classifying an image based on probability and chances an image might have more than one emotions. This limitation highlights the pressing need for more sophisticated approaches capable of accurately modeling emotion mixtures.

The task of recognizing facial expressions is inherently challenging. It becomes particularly intricate when faced with blended expressions, where multiple emotions coexist in a single facial display. Additionally, detecting subtle emotions, like faint smiles, demands a precise approach. Addressing these complexities necessitates the application of advanced techniques, and Convolutional Neural Network emerge as a promising solution.

AIM AND OBJECTIVES

Aim of the project:

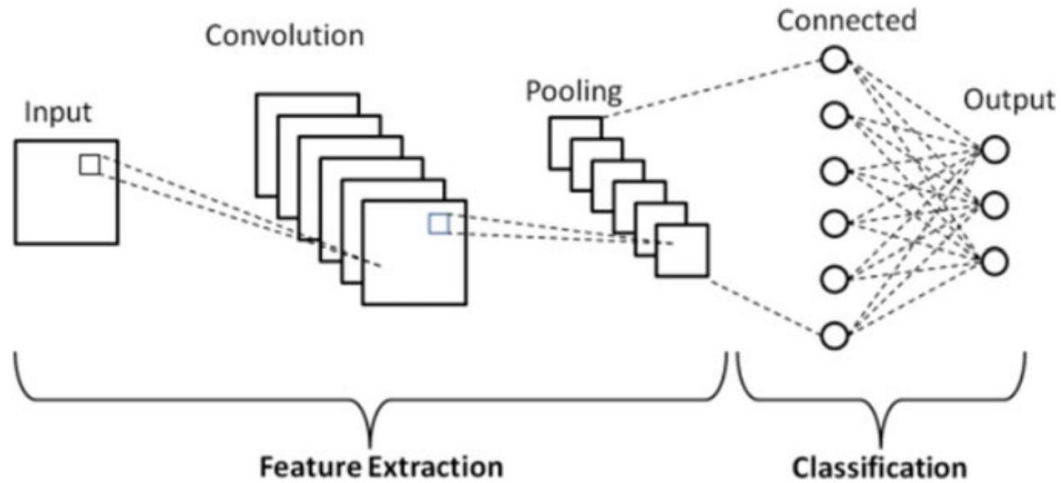
Develop a FER system that shows the list of emotion and probability using CNN that can predict based on eight classes of facial emotions.

SPECIFIC OBJECTIVES

1. Design CNN architecture optimized for probability prediction on 8 classes of facial emotions.
2. Train model on dataset .
3. Evaluate with metrics like micro/macro F1 score, ROC curve and Analyze model for blended emotion insights.

Methodology

Training Stage



Facial Expression
Dataset



Crop Face

Cleaned
Dataset

Fine-
Tuning

Emotion
Recognition
Model

Testing Stage



Test Image

Crop Face



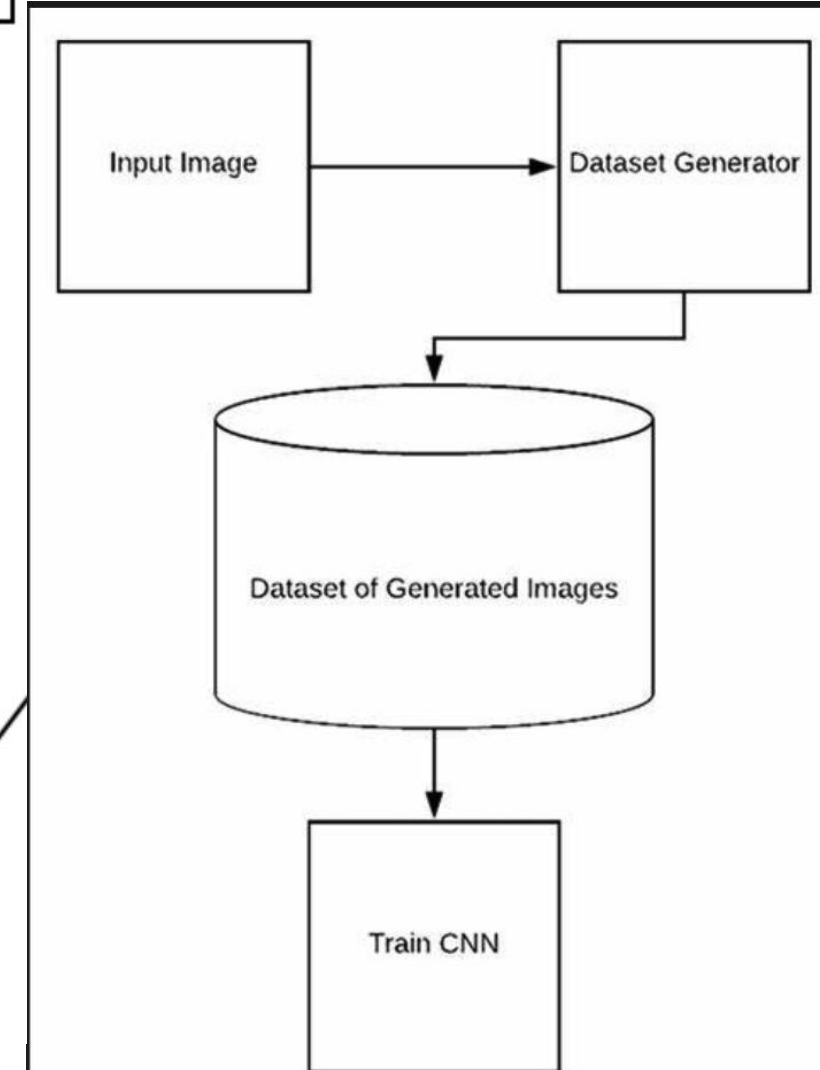
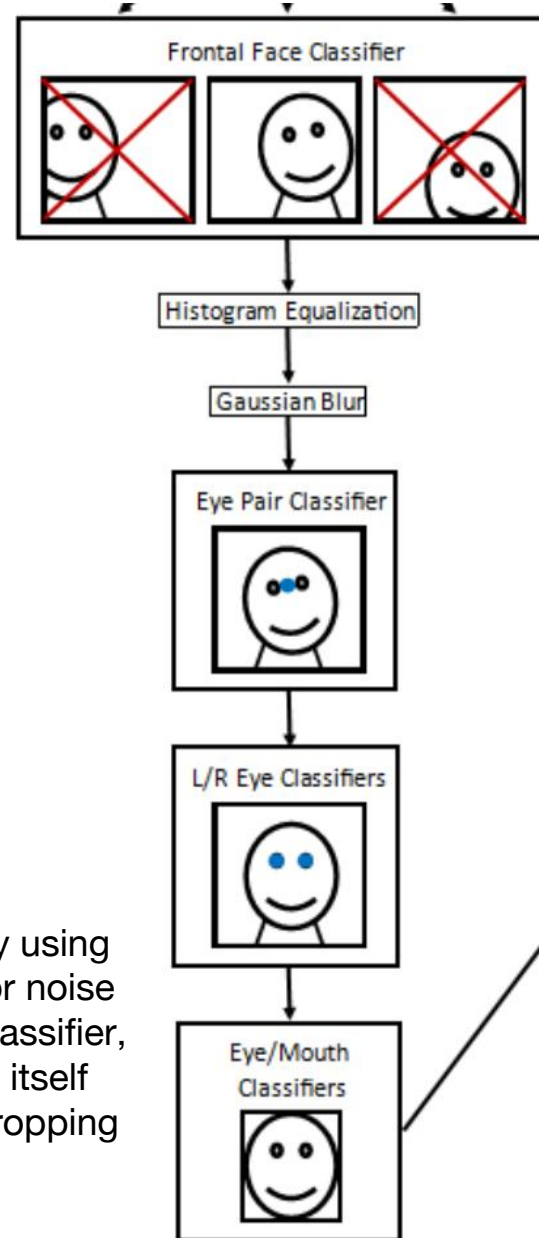
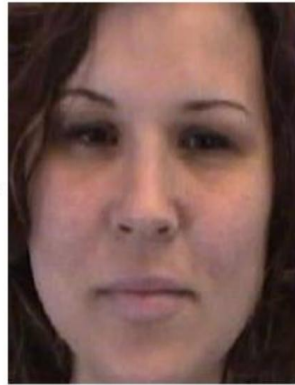
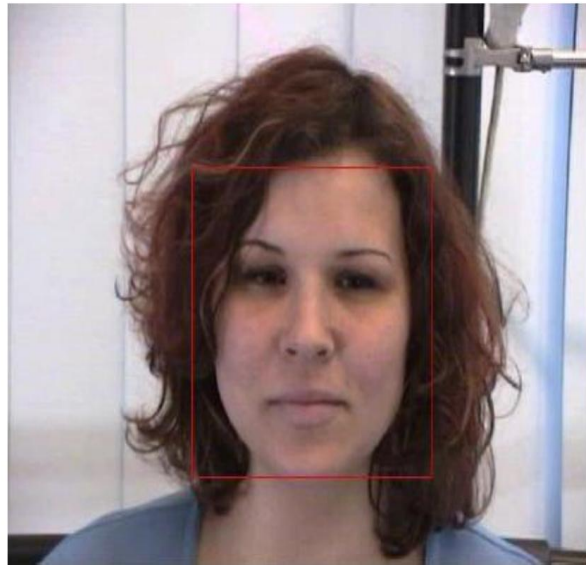
Cropped Face

Emotion
Recognition
Model

Afraid
Angry
Disgusted
Sad
Happy
Surprised
Neutral

Output
Probability
of Each
Emotion

Dataset Preprocessing and Argumentation



The use of OpenCV libraries helps in the Capturing of Facial Images by using histogram equalization for contrast enhancement and Gaussian blur for noise reduction, which makes it easy for the detection of eyes using a eye classifier, by detecting the left and right eye, makes it easier for OpenCV to align itself with the face rotation so as to detect the nose and mouth, for better cropping of the image

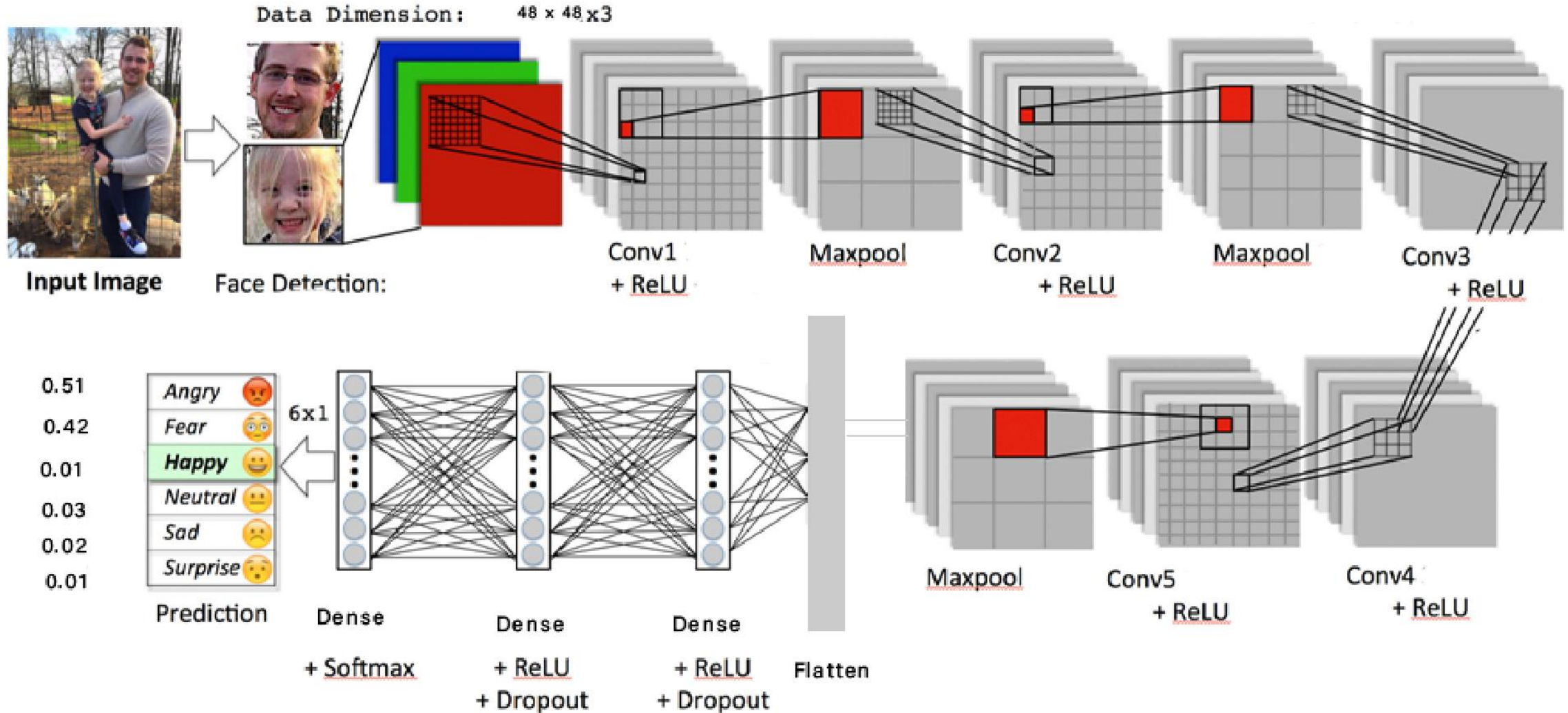
More information about Dataset

The AffectNet dataset has over 27,836 images with 8 classes of images which are :-

- Anger
- Comtempt
- Disgust
- Fear
- Happy
- Neural
- Surprise
- Sad

Each images are of size 96 X 96, which are in the form RGB, which is publicly accessible through Kaggle datasets and research publications, AffectNet empowers researchers and developers to leverage its rich database for facial expression analysis and emotion recognition advancements.

CNN Architecture



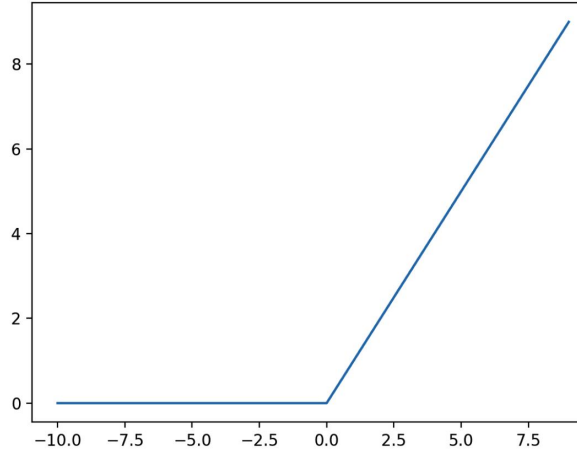
Function used doing Model Compilation

- Activation Function

- Relu

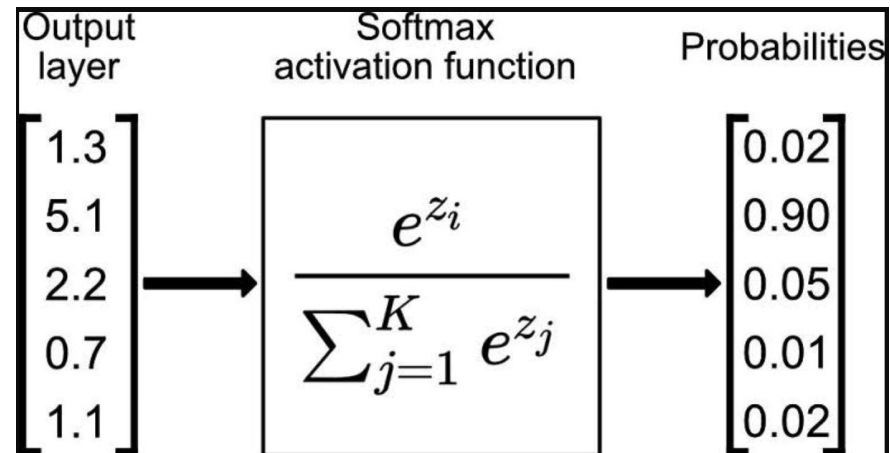
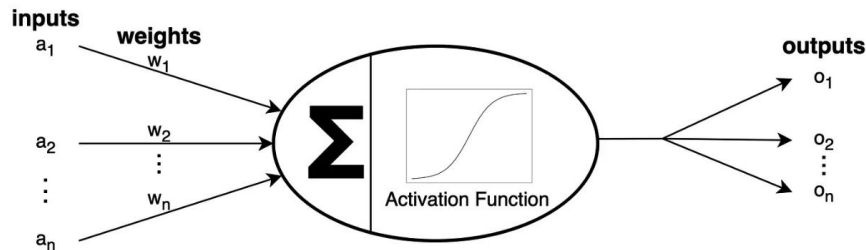
The rectified linear unit (ReLU) is a popular activation function in deep learning and neural networks. It introduces non-linearity and helps with complex pattern recognition. ReLU also solves the vanishing gradients issue and accelerates training convergence.

$$f(x) = \max(0, x)$$



- SoftMax

The softmax activation function is a widely used function that converts raw scores into a probability distribution. It's used in the output layer of neural network models that predict a multinomial probability distribution.



Function used doing Model Compilation

- Loss Function

- `sparse_categorical_crossentropy`

- It is a loss function specifically designed for multi-class classification problems where true labels are provided as integers (e.g., 0 for class A, 1 for class B, etc.) rather than one-hot encoded vectors.
 - It measures the difference between the model's predicted probabilities for each class and the true class label, aiming to minimize this difference during training.

Advantages over Categorical Crossentropy:

1. Efficiency:

It avoids the need to convert integer labels to one-hot encoded vectors, saving time and memory, especially for large datasets or many classes.

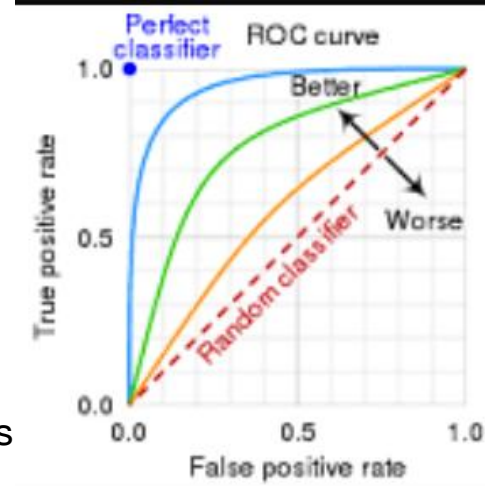
2. Clarity:

It's often more intuitive to provide integer labels directly, making model training and interpretation simpler.

3. Compatibility:

Some deep learning frameworks (like TensorFlow and Keras) offer optimized implementations for `sparse_categorical_crossentropy`, potentially leading to faster training.

Performance and Evaluation



- Performance Metrics:

- a) Micro/Macro F1-score:

F1-score is a harmonic mean of precision and recall, balancing both measures for a more robust evaluation.

$$F1 = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$$

Micro F1-score: Calculates the overall average F1-score across all classes, potentially favoring dominant classes

Equation:

$$\text{Micro F1} = (\text{sum of true positives across all classes}) / (\text{sum of predicted positives across all classes})$$

Macro F1-score: Averages the individual F1-scores for each class, giving equal weight to all emotions.

Equation:

$$\text{Macro F1} = (\text{sum of individual class F1-scores}) / \text{number of classes}$$

- b) ROC Curve:

ROC (Receiver Operating Characteristic) curve visualizes the trade-off between True Positive Rate (TPR) and False Positive Rate (FPR) at different classification thresholds

ROC curve for facial emotion recognition

TPR (also called sensitivity) measures the model's ability to correctly identify true positives (e.g., correctly predicting happy faces).

FPR measures the model's tendency to incorrectly classify negative examples as positives (e.g., classifying a sad face as happy).

AUC (Area Under the Curve) quantifies the overall performance of the model, with higher AUC indicating better discrimination between classes.

EXPECTED CONTRIBUTION TO KNOWLEDGE

- Advance state-of-the-art in multi-label FER.
- Provide insights into blended facial emotion recognition.
- Enable new applications benefitting from facial emotion detection.

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